| Failure mode<br>measured on the VCS | VCS measured evidence<br>proven in this benchmark | Neural-network analogue<br>documented, not proven here | Status<br>here → there |
|---|---|---|---|
| 1 **Gradient vanishes on discrete / index outputs** | **0.000**<br>position-regime F<br>single record<br>Vanilla saliency & IG = 0.000 on the position regime (provably zero at a strobe-timed argmax); whole gradient bucket 0.068. | Shattered / saturated gradients; non-differentiable argmax & index readouts. | **proven here**<br>**analogue there** |
| 2 **Saliency is model-invariant (sanity-check fails)** | **0.000**<br>sanity-check margin<br>single record<br>Program-randomization: saliency cosine = 1.000 while 95% of RAM changed -> sanity check FAILS. | Saliency unchanged under model- & label-randomization; behaves like an edge detector. | **proven here**<br>**analogue there** |
| 3 **Correlation ≠ causation (Granger & tuning trap)** | **0.136**<br>Granger F<br>95% CI [0.02, 0.29]<br>Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116. | Spurious Granger edges; 'present ≠ used' tuning & attention read as explanation. | **proven here**<br>**analogue there** |
| 4 **Single-unit ablation has an interaction blind-spot** | **0.414**<br>connectome F<br>95% CI [0.08, 0.78]<br>Single-unit lesion is faithful (0.987) yet misses the wiring: connectome recovered at only 0.414. | Single-unit ablation misleading; computation is distributed across many units. | **proven here**<br>**analogue there** |
| 5 **SAE/probe: present-not-used** | **0.162**<br>selectivity F<br>95% CI [0.08, 0.24]<br>SAEs match a variable, yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are not causal. | Control-task selectivity gap; superposition / polysemantic neurons. | **proven here**<br>**analogue there** |
| 6 **IG/EG baseline sensitivity** | **0.034**<br>EG F (all-regime)<br>95% CI [0.00, 0.10]<br>IG magnitude is baseline-sensitive (up to 0.999 across games); a content-byte baseline kills IG; EG lands at 0.034. | Attribution depends on the (arbitrary) baseline choice; no canonical reference input. | **proven here**<br>**analogue there** |